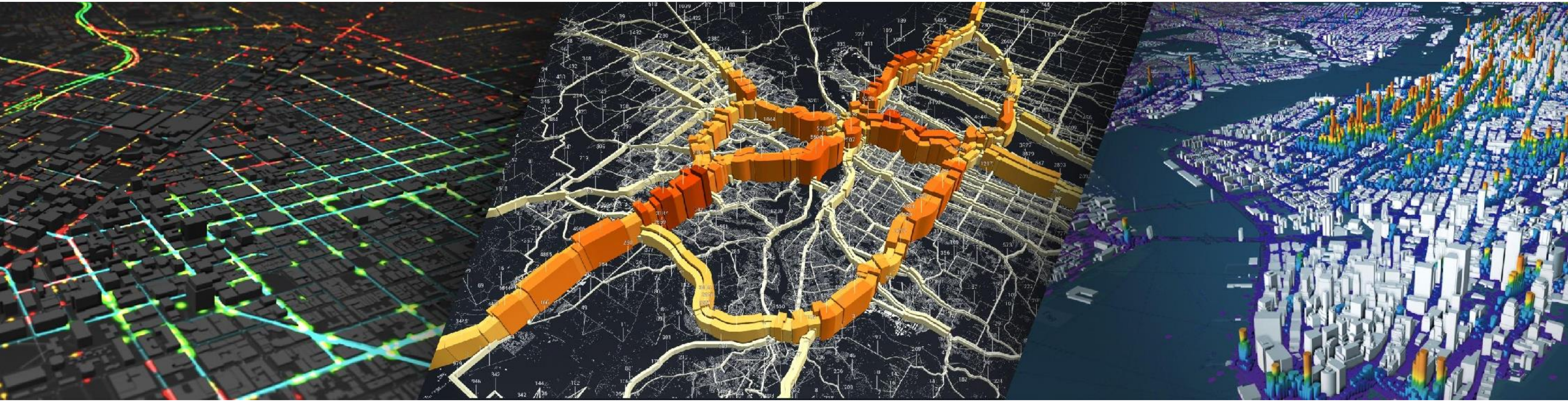
Structural Equation Modeling

SAMPLING SIZE DETERMINATION



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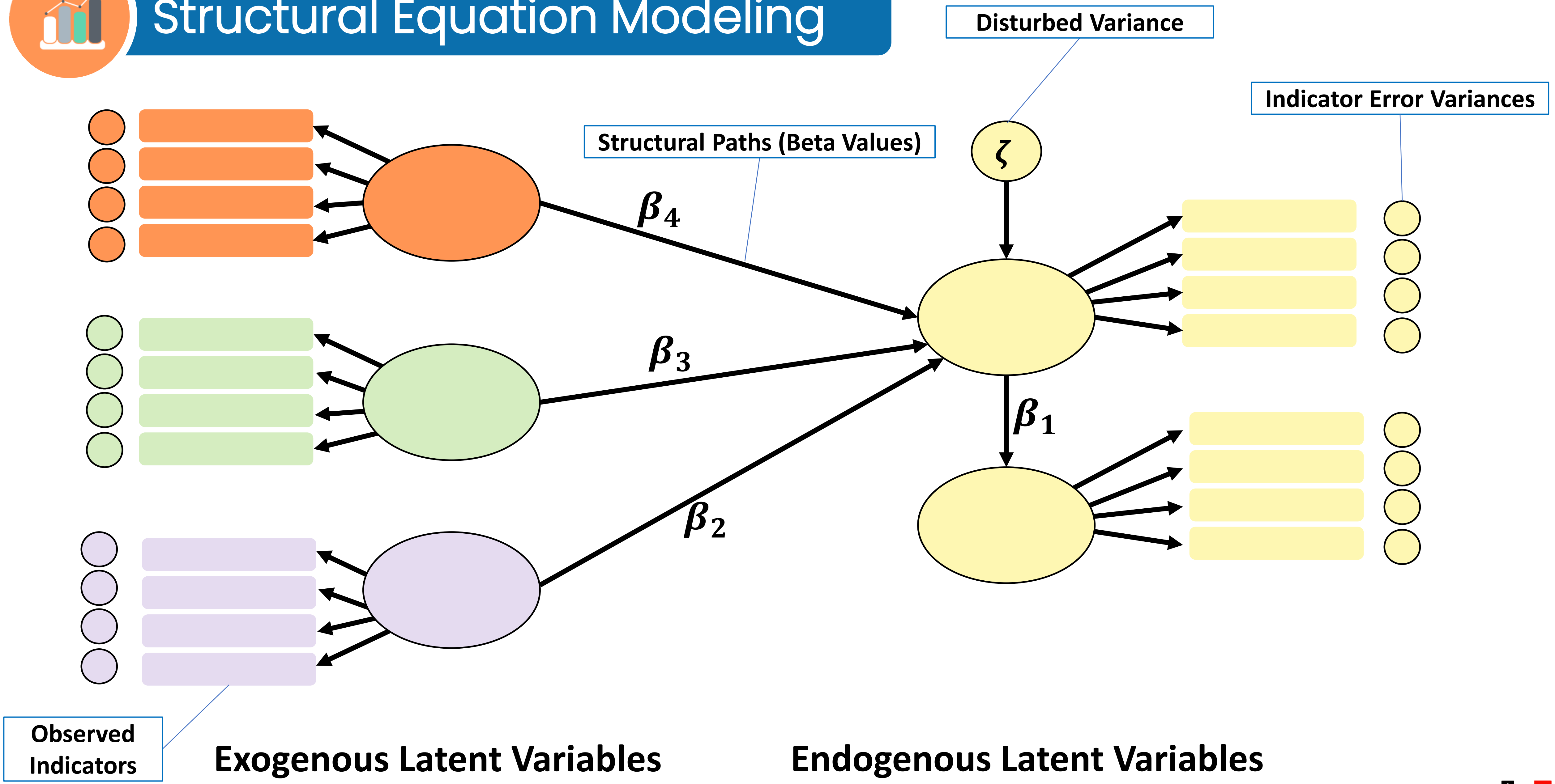
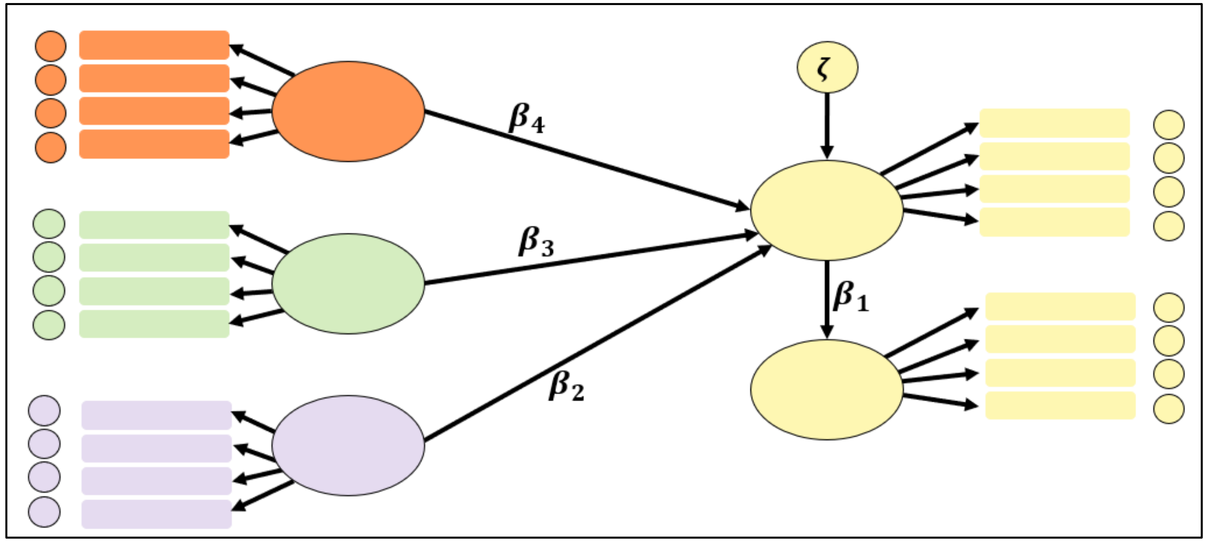
Structural Equation Modeling



Structural Equation Modeling



Number of unique Elements in the observed Covariance Matrix

20 observed indicators

$$\frac{p(p+1)}{2} \qquad \frac{20(20+1)}{2} \qquad 210$$

Factor Loadings

(Latent Variables)(Indicators – 1)

One loading per factor fixed to 1 for identification

(5)(4 – 1) = 15

Indicator Error Variances

20 indicators

Variances of Exogenous Factors

3

Covariances among exogenous latent variables

$$\frac{x(x-1)}{2} \qquad \frac{3(3-1)}{2} = 3$$

Structural Paths

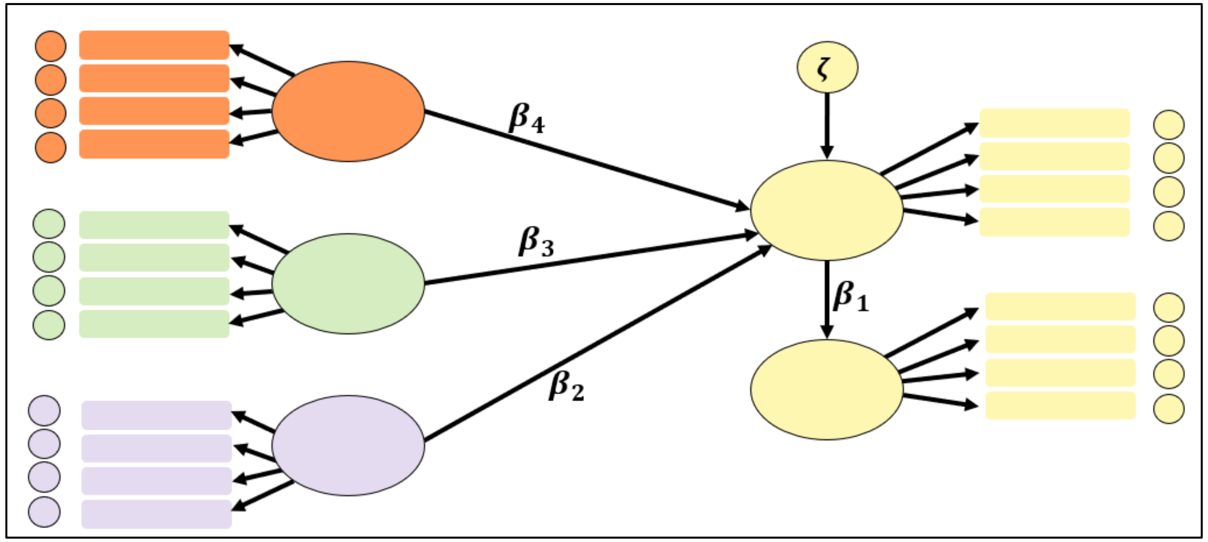
$\beta_1 to\ \beta_4$ $\mathbf{4}$

Disturbance Variances

2

TOTAL = 47 FREE PARAMETERS

Structural Equation Modeling



Number of unique Elements in the observed Covariance Matrix

20 observed indicators

$$\frac{p(p+1)}{2} \qquad \frac{20(20+1)}{2} \qquad 210$$

Degrees of Freedom (df)

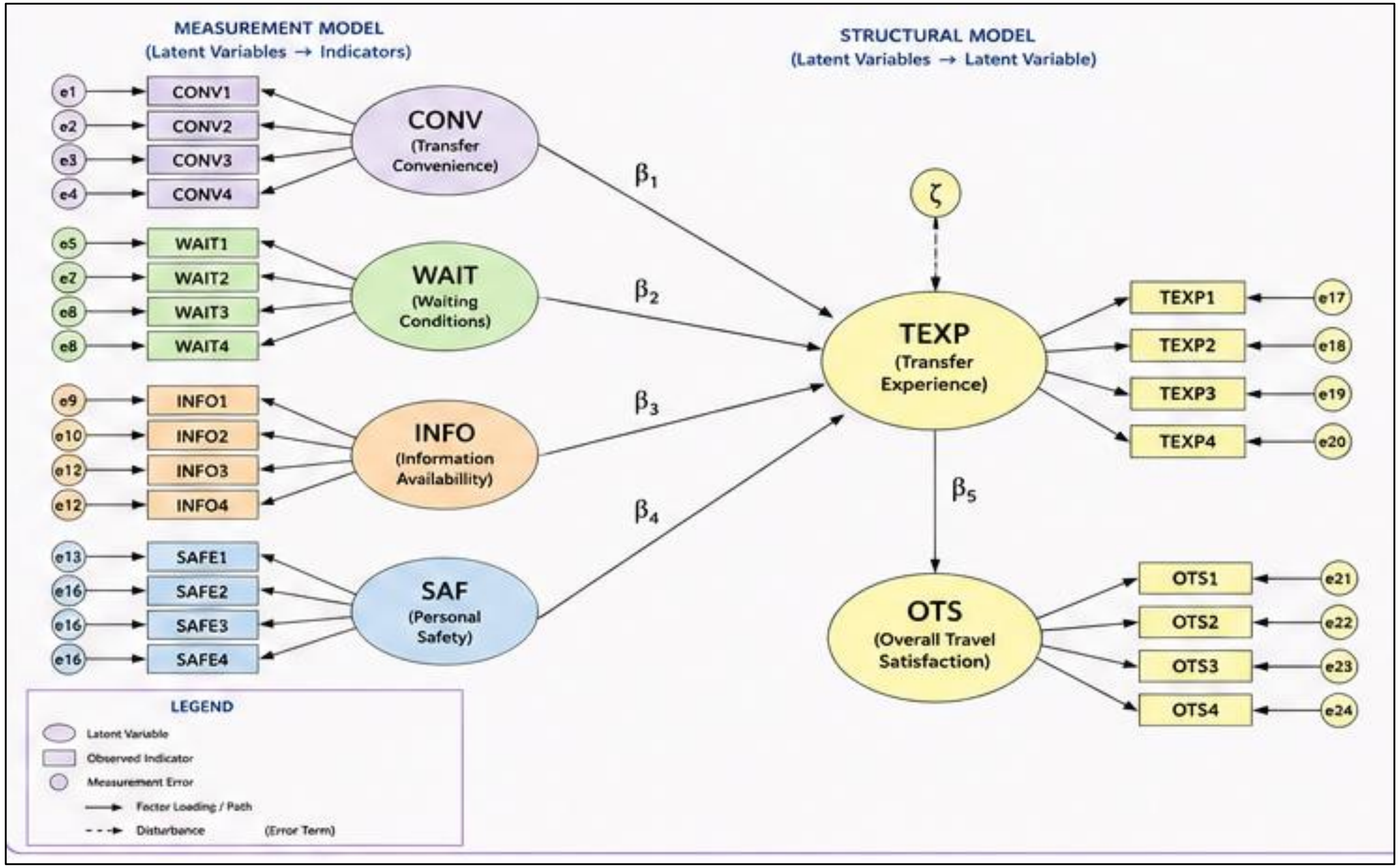
210 unique elements

47 free parameters

df = 210 – 47

$\boxed{\textbf{df = 163}}$

Structural Equation Modeling



24 observed indicators

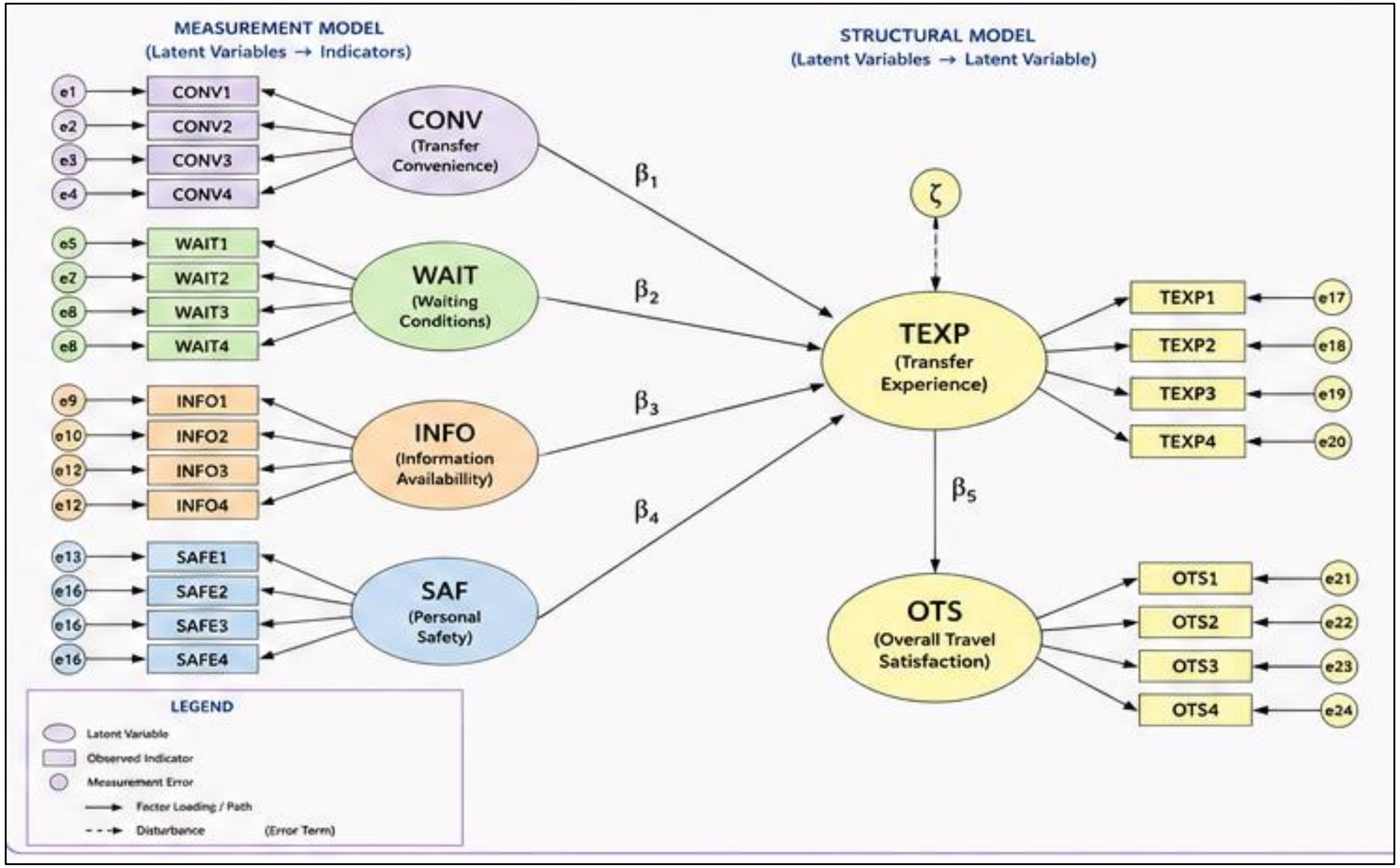
Number of unique Elements in the observed Covariance Matrix

$$\frac{p(p+1)}{2}$$

$$\frac{24(24+1)}{2}$$

$$300$$

Structural Equation Modeling



Factor Loadings

(Latent Variables)(Indicators – 1)

One loading per factor fixed to 1 for identification

(6)(4 – 1) = 18

Indicator Error Variances

24 indicators

Variances of Exogenous Factors

4 (CONV, WAIT, INFO, SAF)

Covariances among exogenous latent variables

$$\frac{x(x-1)}{2} \qquad \frac{4(4-1)}{2} = 6$$

Structural Paths

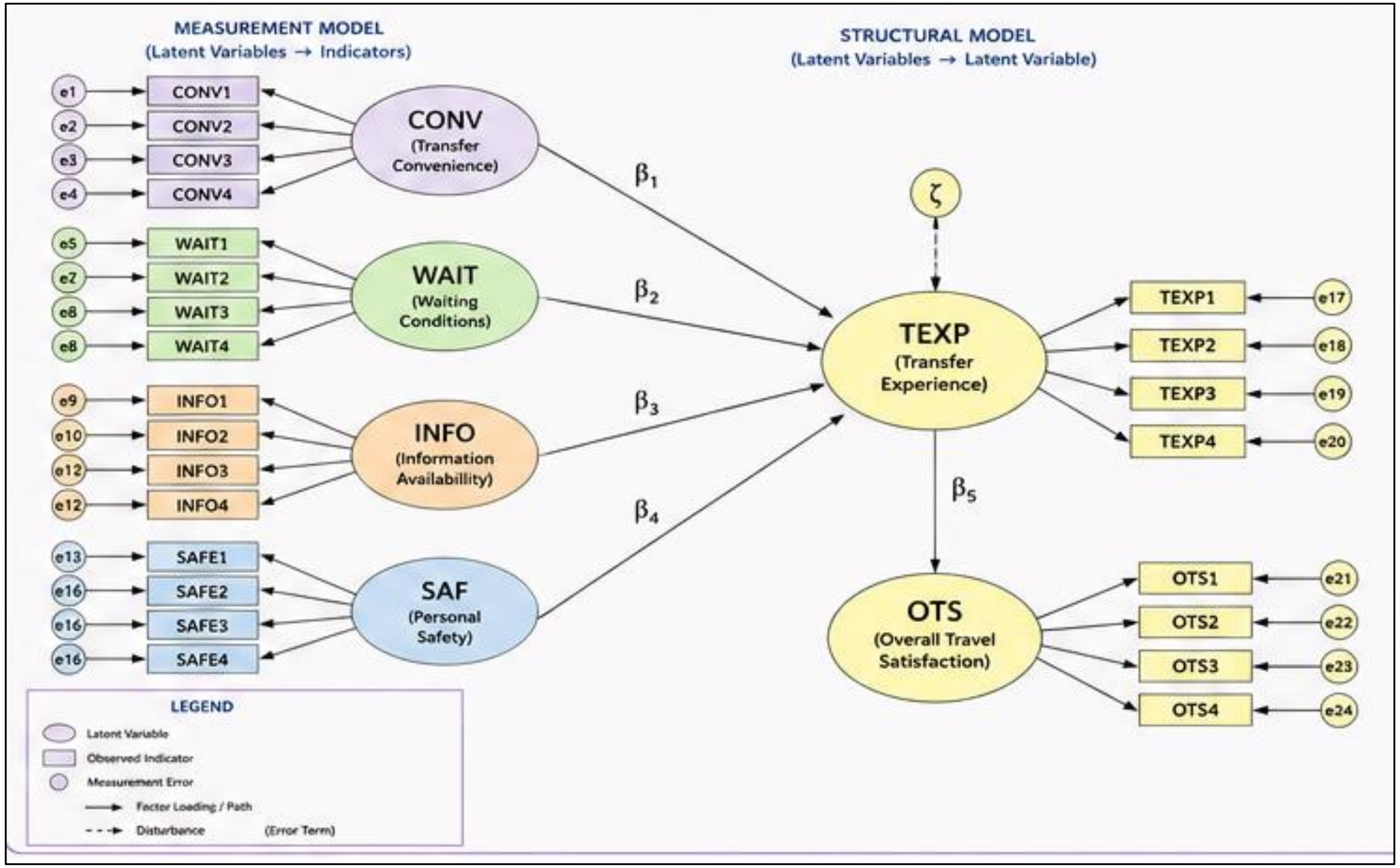
$\beta_1 to\ \beta_5$ **5**

Disturbance Variances

TEXP, OTS **2**

TOTAL = 59

Structural Equation Modeling



Degrees of Freedom (df)

300 unique elements

59 free parameters

df = 241

Structural Equation Modeling

Sample Size Determination Calculator

https://lesterjayollero.github.io/Structural_Equation_Modeling_Sample_Size_Calculator/

Sample Size Calculator

Structural Equation Modeling — Root Mean Squared Error of Approximation (RMSEA)

1. Calculate SEM degrees of freedom

For a full SEM, degrees of freedom are calculated from the number of unique observed covariance elements minus the number of free parameters.

Number of observed indicators (p) — 24
Number of Latent Variables times Number of Statements or Questions per Latent Variable

Free factor loadings — 18
(Latent Variables)(Indicators per Latent Variable - 1)

Indicator error variances — 24

Variances of exogenous latent variables — 4

Covariances among exogenous latent variables — 6
x exogenous factors → x(x-1)/2 covariances.

Structural paths — 5
Number of Beta Arrows

Disturbance variances — 2

Calculate model df

UNIQUE COVARIANCE ELEMENTS
300

TOTAL FREE PARAMETERS
59

CALCULATED MODEL DF
241

2. RMSEA power analysis

Enter the expected RMSEA, significance level, statistical power, and model degrees of freedom.

Expected RMSEA — 0.05
Typical value for close fit: 0.05

Significance level (α) — 0.05
Two-tailed / conventional significance level

Power (1 − β) — 80 %

Degrees of freedom (df) — 13
Calculated below from the SEM measurement model.

Expected Non-response rate — 10 %

Calculate Sample Size · Reset

Results

Required sample size based on the entered RMSEA, α, power, and df.

MODEL DF
13

REQUIRED NON-CENTRALITY
17.847

SAMPLE SIZE, N
551

SAMPLE SIZE WITH NON-RESPONSE RATE
612

The calculation solves for the noncentrality parameter required to attain the requested power, then applies $N = \lambda / (RMSEA^2 \times df) + 1$. The displayed sample size is rounded up to the next whole participant.

Structural Equation Modeling

Sample Size Calculator

Structural Equation Modeling — Root Mean Squared Error of Approximation (RMSEA)

1. Calculate SEM degrees of freedom

For a full SEM, degrees of freedom are calculated from the number of unique observed covariance elements minus the number of free parameters.

1 Inputs

Number of observed indicators (p) Number of Latent Variables times Number of Statements or Questions per Latent Variable	24
Free factor loadings (Latent Variables)(Indicators per Latent Variable - 1)	18
Indicator error variances	24
Variances of exogenous latent variables	4
Covariances among exogenous latent variables x exogenous factors → x(x-1)/2 covariances.	6
Structural paths Number of Beta Arrows	5
Disturbance variances	2

2 Click Calculate

Calculate model df

3 Results

UNIQUE COVARIANCE ELEMENTS
300

TOTAL FREE PARAMETERS
59

CALCULATED MODEL DF
241

Structural Equation Modeling

2. RMSEA power analysis

Enter the expected RMSEA, significance level, statistical power, and model degrees of freedom.

Expected RMSEA Typical value for close fit: 0.05	0.05
Significance level (α) Two-tailed / conventional significance level	0.05
Power (1 − β)	80 %
Degrees of freedom (df) Calculated below from the SEM measurement model.	13
Expected Non-response rate	10 %

Calculate Sample Size | Reset

Results

Required sample size based on the entered RMSEA, α, power, and df.

MODEL DF
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REQUIRED NON-CENTRALITY
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SAMPLE SIZE, N
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The calculation solves for the noncentrality parameter required to attain the requested power, then applies $N = \lambda / (RMSEA^2 \times df) + 1$. The displayed sample size is rounded up to the next whole participant.

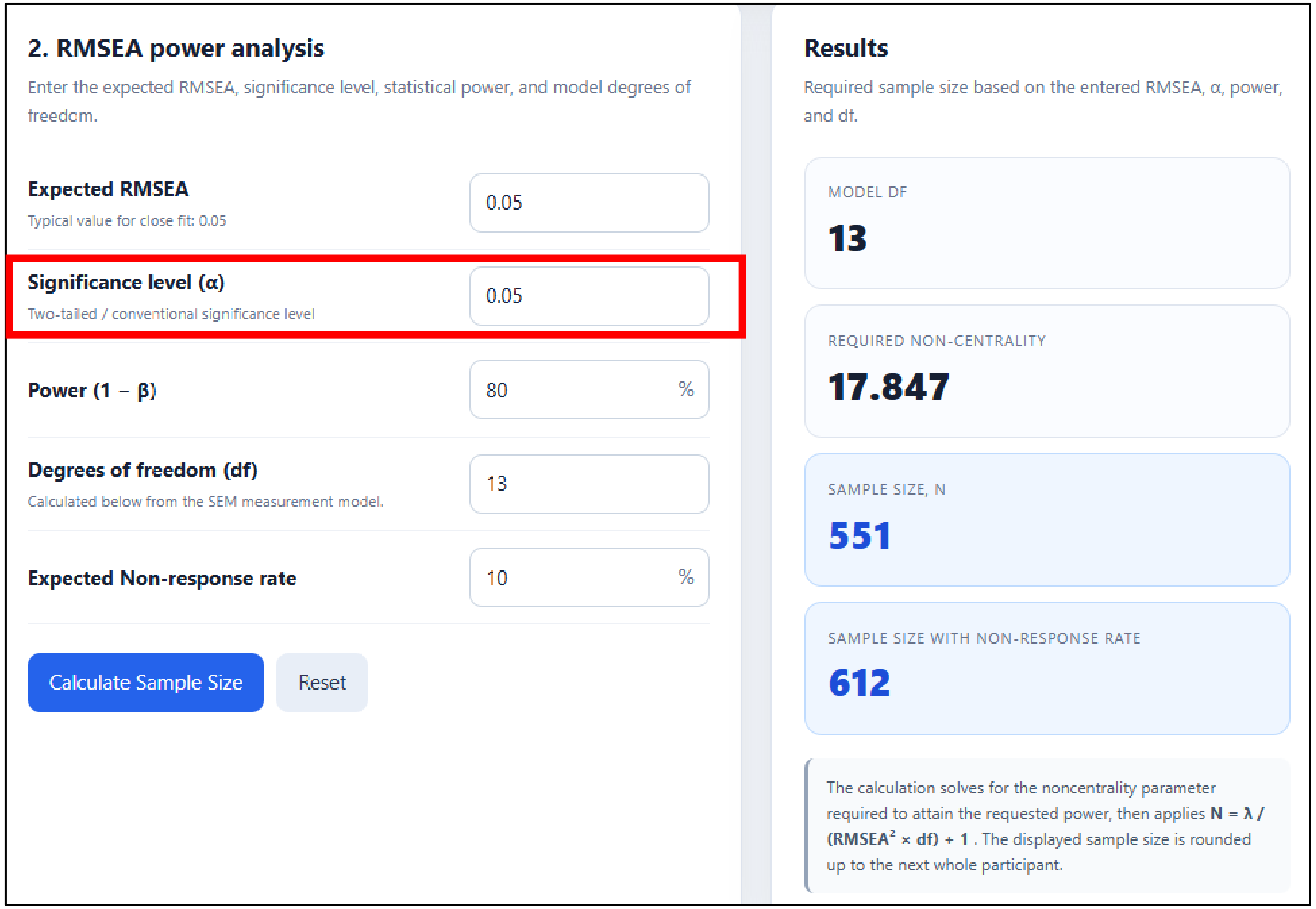
Expected RMSEA

One of the common model-fit indices in SEM

RMSEA	Common interpretation
≤ 0.05	Close/good fit
0.05–0.08	Reasonable/acceptable fit
0.08–0.10	Poor fit
> 0.10	Very poor fit

Sample: Expected RMSEA 0.05

“I expect the SEM model to have an RMSEA of approximately 0.05, and I want enough respondents to detect this level of fit with my chosen statistical power”

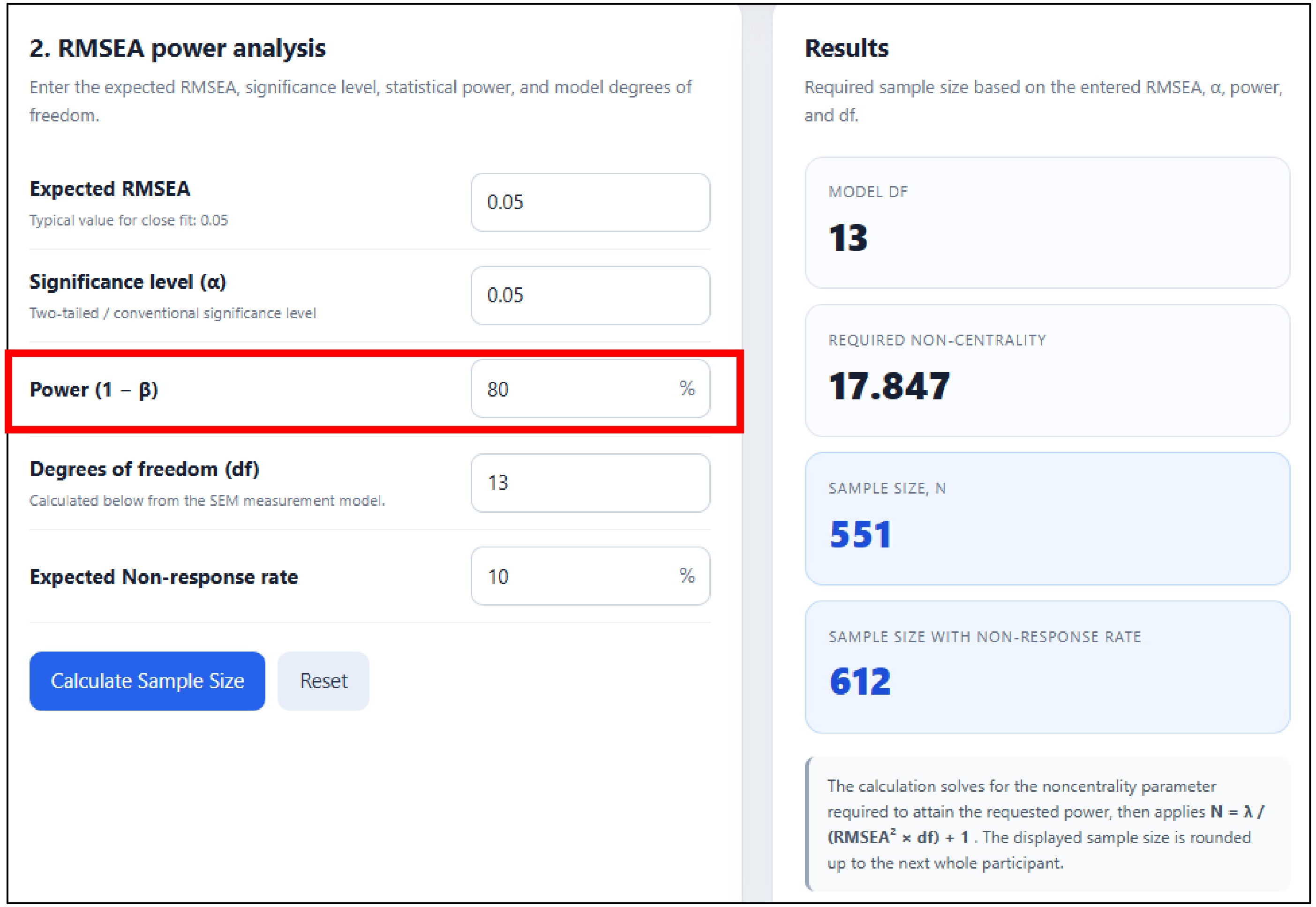


Significance Level

Represents the acceptable probability of making a Type I error.

Threshold you use for deciding whether the statistical evidence is significant.

Structural Equation Modeling



Power

Statistical Power is the probability that your analysis will successfully detect an effect or departure from null hypothesis when effect/departure actually exists.

Sample: 80%

You are designing the study to have an 80% probability of detecting the specified effect under the assumptions of your power analysis.

Higher Power generally requires a Larger Sample

Structural Equation Modeling

Expected Non-response Rate

2. RMSEA power analysis

Enter the expected RMSEA, significance level, statistical power, and model degrees of freedom.

Expected RMSEA
Typical value for close fit: 0.05
0.05

Significance level (α)
Two-tailed / conventional significance level
0.05

Power (1 − β)
80 %

Degrees of freedom (df)
Calculated below from the SEM measurement model.
13

Expected Non-response rate
10 %

Calculate Sample Size
Reset

Results

Required sample size based on the entered RMSEA, α, power, and df.

MODEL DF
13

REQUIRED NON-CENTRALITY
17.847

SAMPLE SIZE, N
551

SAMPLE SIZE WITH NON-RESPONSE RATE
612

The calculation solves for the noncentrality parameter required to attain the requested power, then applies $N = \lambda / (RMSEA^2 \times df) + 1$. The displayed sample size is rounded up to the next whole participant.

Sample: 10%

You expect that approximately 10% of people you recruit will not respond, withdraw, submit incomplete questionnaires, or provide unusable responses.